

9 (**short**) The Young's modulus of composite materials may be estimated using lower bound and upper bound formulae, given on p.5 of the Materials Databook.

(a) Explain the key assumptions behind the derivation of these formulae, and indicate on a sketch how they would apply to a unidirectional fibre-reinforced material. [2]

(b) A layer of unidirectional fibre-reinforced material contains 50% of fibres with modulus E_f in a matrix of modulus E_m , with $E_f/E_m = 100$. A composite panel is manufactured from an even number of layers of this unidirectional material stacked and stuck together. Fibres in alternate layers are oriented at 90° to one another. Derive an expression for the Young's modulus in either of the two directions parallel to sets of fibres. [5]

(c) The layered composite panel in part (b) is manufactured using drawn polymer fibres ($E_f = 120$ GPa) in a thermoset matrix ($E_m = 1.2$ GPa). Calculate the Young's modulus in either of the two directions parallel to sets of fibres. The density of the composite is $\rho = 950 \text{ kg m}^{-3}$. Using the Young's modulus - density property chart in the Materials Databook, identify the nearest material that competes with this composite. [3]